

Varun Ghat Ravikumar

AI and Linguistics enthusiast

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AI Engineer bridging Linguistics and NLP. Specialized in architecting end-to-end LLM pipelines, Knowledge Graphs, and document intelligence. Currently building multi-lingual data extraction pipelines, with experience in video processing.

EDUCATION

Master's in Informatics (Major: AI, Minor: DS)

SEP 2023 — MAR 2026

University of Zurich

Grade: 5.2 / 6.0

Bachelor of Engineering in Computer Science & Engineering

AUGUST 2017 — MAY 2021

JSS Science & Technology University (Formerly SJCE), Mysuru

CGPA: 9.01

EXPERIENCE

PROGRAMMER (CONTRACT) - ISLE / UNIVERSITY OF ZURICH (Part-time)

FEB 2026 — Present

Python, LLM, OCR

Zurich, Switzerland

- Architected an LLM-driven OCR pipeline to automate data extraction and enhancement for diverse word-list PDF books.
- Working on semi-automated IPA creation from orthography and semantic word classification.

RESEARCH ASSISTANT (DDIS GROUP) / UNIVERSITY OF ZURICH (Part-time, 20%)

APR 2024 — DEC 2024

Python, Knowledge Graphs, LLM, GenAI

Zurich, Switzerland

- Developed an end-to-end pipeline for knowledge graph synthetic dataset generation and augmentation, optimized data ingestion to handle 100 million+ entities efficiently. Implemented synthetic data augmentation in text and images using GenAI

ASSOCIATE DATA SCIENTIST & INTERN / AMAGI MEDIA LABS

JAN 2023 — AUG 2023

Python, Video processing, NLP, PyTorch, AWS

Bangalore, India

- **Fast-tracked from Intern to full-time Data Scientist in just 3 months for high-impact project execution.**
- Engineered a POC sports highlight detection system to automate key moment extraction in soccer videos at 6x real-time speed.
- Deployed a multi-modal genre classification model to enhance categorization in a live production environment.
- Enhanced production pipelines by implementing model drift detection metrics.

INTERNSHIP / KERNEL INSIGHTS

NOV 2022 — JAN 2023

Stable Diffusion (GenAI), PyTorch

Bangalore, India

- Fine-tuned Stable Diffusion for consistent storybook stylization. Explored latent seed strategies to control subject poses.

INTERNSHIP / INDIAN INSTITUTE OF ASTROPHYSICS

SEP 2021 — MAY 2022

C++, C, MPI, PLUTO code

Bangalore, India

- Migrated C++/MPI modules to PLUTO to implement data-driven MHD simulations of solar flares, studying CME formation.

PUBLICATIONS

- **Making Grammar Books Generative: A Factorial Study of Synthetic Data for Low-Resource Machine Translation** (Under Review)
V. G. Ravikumar, et. al. | Submitted to ACL Rolling Review (ARR), March 2026
- **Alan's Speakeasy - An Ecosystem for the Evaluation of Conversational Agents**
L. Rossetto, C. Sarasua, O. Inel, V. G. Ravikumar, et al.
Proceedings of the ACM Web Conference 2025 (WWW '25), Companion Volume. [DOI: 10.1145/3701716.3715165]

PROJECTS

- **Low-Resource Language Translation using Synthetic Data from Grammar Books (LLMs, Python, GCP Vertex AI)** - Master's Thesis
Designed an automated OCR + NLP pipeline to extract linguistic rules from grammar books. Generated 190k+ synthetic translation pairs via Gemini. Fine-tuned LLMs on GCP Vertex AI and evaluated 96 experimental setups, outperforming zero-shot baselines.
- **AI Data Analysis Dashboard** | Hack4Good & Caritas (Ongoing)
Developing an internal analytics tool allowing non-technical employees to visualize and analyze CSV/Excel data via natural language queries. Architecting the backend utilizing LLMs and Retrieval-Augmented Generation (RAG) to automate data interpretation.
- **Diffusion Models Research Project (PyTorch)** - Master Project
Explored the inner workings of diffusion models by experimenting with diffusion, cold diffusion and implementing a cold latent diffusion model to analyze learning patterns.

SKILLS

Programming Languages	Python, C, C++, SQL, Javascript
Core AI & ML	PyTorch, Scikit-learn, Transformers, Spacy, LLMs, RAG, Agentic AI
Computer Vision	OpenCV, Diffusion Models, Vision Language Models, OCR
Tools	Docker, Git, Node.js, LangChain, LangGraph, MLflow, FastAPI, Linux
Cloud Platforms	AWS, Azure Machine Learning, GCP (Vertex AI)
Spoken Languages	English (Native), German (B1.1), Kannada (Native), Hindi (Fluent), French (Basic), Chinese (Basic)

CERTIFICATIONS

Machine Learning Engineering Professional Certificate (Google Cloud)	Google Cloud, Coursera
IBM RAG and Agentic AI Specialization	IBM, Coursera
Microsoft Azure for AI and Machine Learning	Microsoft, Coursera
Deep Learning Specialization	DeepLearning.AI, Coursera

ACHIEVEMENTS

Winner - Best Visualization, AI + Environment EcoHackathon	
ETH BioDivX, GainForest, ETH AI Center, 2024.	
Winner - Coding Competitions, JSS Science & Technology University	
Secured 1st and 2nd place in various competitive coding and problem-solving events.	
Organizer - Science and Tech Quizzes	
Organized tech and science quizzes JSSSTU Mysore and NPS Mysore.	

INTERESTS

Creative Tech: Blender, Unity 3D, Photoshop	
Other: Astrophysics, Music, Sketching, Bouldering, Language learning	